

Q1. Which of the following is a relation?

- (A) $\{(1, 2), (2, 3), (3, 4)\}$
 (B) $\{(1, 1), (2, 2), (3, 3)\}$
 (C) $\{(1, 2), (1, 3), (1, 4)\}$
 (D) $x^2 + y^2 = 4$
 (E) $y = 2x + 1$

Q2. What is the domain of the function $f(x) = \sqrt{x}$?

- (A) $(-\infty, 0]$
 (B) $[0, \infty)$
 (C) $(-\infty, \infty)$
 (D) $[0, 10]$
 (E) $(-10, 0)$

Q3. Determine if the relation $y = x^2$ is a function using the vertical line test.

- (A) Yes, it is a function.
 (B) No, it is not a function.
 (C) Maybe, it depends on the domain.
 (D) Only for $x > 0$
 (E) Only for $x < 0$

Q4. Find the range of the function $f(x) = |x|$.

- (A) $(-\infty, 0]$
 (B) $[0, \infty)$
 (C) $(-\infty, \infty)$
 (D) $(-10, 10)$
 (E) $[0, 1]$

Q5. What is the value of $f(-2)$ if $f(x) = 2x + 1$?

- (A) -3
 (B) -1
 (C) 1
 (D) 3
 (E) 5

Q6. Which of the following is an example of function notation?

- (A) $y = 2x + 1$
 (B) $f(x) = 2x + 1$
 (C) $x = 2y + 1$
 (D) $y = x^2$
 (E) $x = y^2$

Q7. Determine the domain of the function $f(x) = \frac{1}{x}$.

- (A) $(-\infty, 0]$
 (B) $[0, \infty)$
 (C) $(-\infty, \infty)$
 (D) $(-\infty, 0) \cup (0, \infty)$
 (E) $[0, 1]$

Q8. What is the range of the function $f(x) = \frac{1}{x}$?

- (A) $(-\infty, 0]$
 (B) $[0, \infty)$
 (C) $(-\infty, \infty)$
 (D) $(-\infty, 0) \cup (0, \infty)$
 (E) $[0, 1]$

Open Ended Questions (FRQ)**Q9.** Graph the function $f(x) = x^2 - 4x + 3$ and find its domain and range.**Q10.** Given the function $f(x) = |x - 2|$, evaluate $f(-2)$, $f(0)$, and $f(2)$, and determine the domain and range of the function.**Chef's Tips**

- Tip 1: Make sure to read each question carefully.
- Tip 2: Use proper notation and show all work.
- Tip 3: Use the vertical line test to determine if a relation is a function.